

10 (a) Striated muscle is composed of myofibrils. Myofibrils contain several structural proteins including troponin, tropomyosin, actin and myosin.

Outline the roles of these four structural proteins in the contraction of a sarcomere.

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(b) Drugs that cause muscle paralysis (paralytic drugs) are used during surgery to stop the patient moving. One commonly used paralytic drug is succinylcholine, which works by preventing contraction of muscles. Succinylcholine is able to prevent muscles contracting because it has a similar shape to acetylcholine.

Suggest how succinylcholine is able to prevent muscles contracting.

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- (c) Duchenne muscular dystrophy (DMD) is a genetic disease that is caused by a single gene. DMD affects striated muscle, and symptoms of the disease first appear at an early age.

A fibrous protein, dystrophin, stabilises muscle fibres during contraction. A person with DMD produces non-functioning dystrophin or no dystrophin at all. The disease occurs in about four in 100 000 people and mainly affects boys.

Suggest **and** explain why boys are more likely to have DMD than girls.

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